Philosophy of Science

Faculty of Science, University of Amsterdam

Retake Final Exam

7 December 2021, 18:00-20:00

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The exam was written by Sebastian De Haro and peer-reviewed by David van den Berg

INSTRUCTION

1. Write your name on every separate sheet of paper you hand in.
2. Make sure that your handwriting is legible.
3. Formulate your answers clearly and accurately, in grammatically correct sentences.
4. Give concise answers: use at most half a page for each question.
5. You are not allowed to leave the examination room during the first 15 minutes or during the final half hour of the exam (i.e. no re-entering).
6. Please switch off any electronic devices and put them in your bag.
7. Please number your pages and hand them in showing the questions in the correct order.

Question	
1.	**10**
2.	**10**
3.	**10**
4.	**10**
5.	**10**
6.	**10**
7.	**10**
8.	**10**
Total	**80**

QUESTIONS

1. Explain briefly the verifiability criterion of meaning, and its significance for the problem of demarcation.

A statement is meaningful (scientific) if it is verifiable, i.e. its truth can in principle be determined through observation or experiment. ('The meaning of a statement is its method of verification'). It solves the problem of demarcation because theories whose statements are not verifiable (pseudo-statements) are pseudo-science: they seem to say something (like metaphysics), but they are literally meaningless.

2. Imre Lakatos analyses the development of science in terms of 'scientific research programmes'. Describe the components of research programmes, and explain what, according to Lakatos, scientific progress consists in.

(1) Hard core; (2) protective belt with auxiliary hypotheses; (3) positive heuristic; (4) negative heuristic. Progressive programme: leads to new discoveries. Degenerating programme: only accommodates known facts.

3. Popper praised Albert Einstein's theory of general relativity as a truly scientific theory, citing the example of Eddington's expedition. This expedition set out to measure the shift of a distant star during a solar eclipse. Einstein's theory states that large masses, like the sun, curve space-time. Thus, incoming light rays will follow a curved path around the sun and will appear in a different spot as seen from the earth. The theory makes a very precise prediction about this shift. In 1953, however, Einstein was asked how he would have responded to a measurement outcome that did *not* show this shift: 'Then I am very sorry for the dear Lord [Eddington], for the theory is correct.' Would, and-or could, this be acceptable for Popper? Explain your answer.

No, assuming that the measurement was correct: the theory is falsified and has to be rejected. Einstein shows a dogmatic attitude. However: perhaps the measurement was not correct. Given the Duhem-Quine thesis, scientists can always put blame elsewhere and retain the theory/hypothesis. However, according to Popper, this is only acceptable if there are independent reasons for rejecting the measurement result (no ad hoc hypothesis).

4. Thomas Kuhn compares scientific revolutions, which consist in a change of paradigm, to visual 'Gestalt switches', in which 'what were ducks in the scientists' world before the revolution are rabbits afterwards'. Explain why Kuhn makes this comparison.

Two reasons: it's a radical shift, a conversion experience that suddenly makes one see the world in a different light; and it relates to his view on observation, inspired by the Gestalt psychologists, which is 'theory-laden'.

5. What does Helen Longino mean by the statement that objective scientific knowledge has a 'social character'? How does something become 'scientific knowledge'? Give some examples of social mechanisms that secure objectivity.

Through the participation of two or more individuals, there is a process of critical emendation and modification of the individual processes. Each of them are transformative. Examples: replication of experiments with variations, peer review that brings another point of view on the phenomena (so the authors interpreted the phenomena free of subjective preferences), treatment after publication (refining of new ideas and techniques), absorption into knowledge (subsequent citation, use and modification by others). (Curd & Cover pp. 175-176)

6. The under-determination thesis was first formulated by Pierre Duhem, and it was later on strengthened by W.V.O. Quine. State this thesis in Duhem's own version, and explain it with an example.

Given that a set of premises {P1, P2,…,Pn} deductively entails a statement O describing a possible observation, and that from experimental observation one concludes that the statement O' is true, where O' entails that O is false, it follows that the conjunction of the premises is false. It does not follow that any particular premise is false. Also: there is no crucial experiment to choose between two hypothesis in an irrefutable way, because hypotheses can only be tested in combination with auxiliary assumptions.

7. Carl Hempel's 'deductive-nomological model' for a long time was the "received view" of scientific explanation. Briefly summarize Hempel's model, and give at least one argument *against* the model's being a satisfactory account of scientific explanation.

D.N.=Deductive argument in which the explanandum (phenomenon to be explained) is deduced from premises containing general laws and particular facts and (initial/boundary) conditions (the explanans).

Objections: explanations do not always involve laws (causal relevance could be enough), asymmetry of explanations vs. symmetry of deductions, irrelevance (may add irrelevant information), apparently missing causation (correlation or expectation ≠ explanation): barometer drop example.

8. One of the arguments discussed by van Fraassen and Musgrave, in their debate on scientific realism, is about scientific explanation. Van Fraassen attacks the realist demand for explanation, and argues that explanation is not one of the aims of science. Musgrave then responds that van Fraassen tacitly conflates 'realism with essentialism, … the demand for explanation

with the demand for ultimate explanation'. What does Musgrave mean by this, and how is it a reply to van Fraassen?

Ultimate scientific explanations serve to remove puzzlement. Non-ultimate scientific explanations do not serve this pragmatic function of removing puzzlement (they relocate it and enhance it), yet they are real explanations. In other words, there can always be better or deeper explanations, and scientific theories certainly do aim to give explanations, but there is no unlimited demand for explanations. (pp. 1102-1103)